Hemanth.S

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Education

Vellore Institute of Technology — Vellore
Bachelor of Information Technology — CGPA: 7.35 — *2023 – 2027*

Riverside Public School — Kotagiri
Grade X- 78 percent —— Grade XII- 78 percent

SUMMARY

- Enthusiastic B.Tech Information Technology student with interests in Database Systems, Machine Learning, Artificial Intelligence, and Blockchain. Skilled in scalable application development and AI-driven solutions. Completed a one-month internship in Python Application Development at I-Net Secure Labs Pvt Ltd. (June 2026).

Experience

I-Net Secure Labs Pvt. Ltd. — Chennai
Internship Training in Python Application Development — *Jun 2026*

- Successfully completed internship training from 01-06-2026 to 30-06-2026.
- Gained valuable practical knowledge through the Tech Team during the internship training.

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | *Oracle* — Jun 2026

Projects

StreamGuard-XAI: Real-Time Fake Job Detection | *Python, Apache Kafka, Apache Spark, LightGBM*

- Built a real-time pipeline that streams job postings through Apache Kafka and Spark to flag fraudulent listings.
- Trained a cost-sensitive LightGBM model with 16 engineered fraud signals: 88.48% F1 vs. 61.04% for a baseline.
- Added explainable AI for every verdict and reached 1.18 ms per prediction, 28.6x faster than Random Forest.

Multimodal Emotion Recognition (Deep Learning) | *Python, PyTorch, RoBERTa, DeBERTa*

- Built a model that combines text, audio and video to recognize 7 emotions in conversations (MELD dataset).
- Designed cross-attention fusion with a network that learns how much to trust each modality per utterance.
- Achieved 67.89% accuracy and 66.30% weighted F1, comparable to published state-of-the-art results.

Online Auction Platform | *React.js, Node.js, Express.js, MongoDB, Socket.io*

- Built a full-stack online auction platform using React.js for the frontend and Node.js with Express.js for backend services with MongoDB as the database.
- Integrated JWT-based authentication to manage user login, authorization, and secure session handling.
- Implemented real-time bidding with Socket.io, with race-safe bid processing and automatic auction closing.
- Added an AI price predictor (ridge regression) that estimates final selling prices with an 80% range.

Mining on Public Speeches (Machine Learning / NLP) | *Python, TextBlob, VADER, SBERT, Scikit-learn, Matplotlib*

- Developed an NLP-based system to analyze sentiment and emotional tone from more than 1400 public speeches.
- Applied TextBlob and VADER for sentiment polarity detection and detailed sentiment classification.
- Used Sentence-BERT embeddings to perform semantic clustering and identify key thematic patterns in speeches.
- Conducted trend analysis and visualization to extract insights from large-scale textual data.

SecureShield: Web Application Firewall | *Python, FastAPI, React, SQLite, JWT*

- Built a Layer-7 WAF in FastAPI that inspects requests for SQLi, XSS, path traversal, command injection and scanner bots.
- Scored each request 0-100 to block, flag or allow it, with custom regex rules that can be edited while it runs.
- Added a sliding-window rate limiter that blacklists repeat offenders, plus JWT and bcrypt authentication.
- Built a React security dashboard with an attack sandbox for testing exploit payloads; covered by 11 pytest tests.

Technical Skills

Languages: Java, Python, SQL (Postgres), JavaScript, HTML/CSS
Frameworks: React, Node.js, Express.js, Socket.io, Flask, FastAPI, Apache Spark, Apache Kafka
Developer Tools: Git, Docker, VS Code
Libraries: pandas, NumPy, Matplotlib, scikit-learn, PyTorch, Hugging Face Transformers, LightGBM